

Yijin Zeng

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SUMMARY

Final year PhD candidate in Statistics and Machine Learning at Imperial College London.

- Research focuses on developing new methodologies for statistical hypothesis testing, statistical correlation, and change point detection in the presence of missing data.
- Applications include A/B testing, causal inference, information retrieval, and continual learning.
- Strong statistics and machine learning theoretical foundations with hands on experience in end-to-end data science, machine learning, and deep learning projects.
- Proficient in Python and R. Solid understanding of data structures and algorithms.

EDUCATION

- **Imperial College London**, London, UK Oct. 2021– Oct. 2025 (Estimated)
PhD in Statistics and Machine Learning ([StatML CDT](#), Imperial & University of Oxford)
 - *Fully funded by Roth Scholarship*
- **Imperial College London**, London, UK Oct. 2020 – Sep. 2021
MSc in Statistics
 - *Degree classification: Distinction*
- **Beijing Institute of Technology**, Beijing, China Sep. 2016 – Jun. 2020
BSc in Mathematics
 - *Top 10% of cohort; Thesis with Distinction*

WORK EXPERIENCE

- **TESCO Data Scientist PhD Internship (London)** Aug.2024 – Oct.2024
 - Worked with forecasting team and supply chain team, delivering an end-to-end machine learning pipeline for demand forecasting, including feature selections, feature engineering, data cleaning, pre-processing, model training, and evaluation.
 - Designed and trained multiple deep neural network architectures for supply chain forecasting, achieving accuracy comparable or higher to TESCO's current approach with less workflows.
- **IQVIA Data Analyst Internship (Beijing)** Jul.2019 – Sep.2019
 - Cleaned and analyzed survey and experimental datasets, performing feature engineering, missing-data handling, and statistical testing for supporting final conclusions.
 - Worked on data visualization for efficiently communicating numerical results to managers and non-technical stakeholders.

PUBLICATIONS

- **Zeng Y**, Adams NM, Bodenham DA. On two-sample testing for data with arbitrarily missing values. arXiv preprint [arXiv:2403.15327](https://arxiv.org/abs/2403.15327). 2024 Mar 22.
 - *A two-sample hypothesis testing method for partially observed univariate data to detect location shifts, with no assumption of the values of missing data.*
- **Zeng Y**, Adams NM, Bodenham DA. MMD Two-sample Testing in the Presence of Arbitrarily Missing Data. arXiv preprint [arXiv:2405.15531](https://arxiv.org/abs/2405.15531). 2024 May 24.

- *Extended our framework for high-dimensional data with any distributional shifts.*
- **Zeng Y**, Adams NM, Bodenham DA. Exact Bounds of Spearman’s footrule in the Presence of Missing Data with Applications to Independence Testing. arXiv preprint arXiv:2501.11696. 2025 Jan 20.
 - *Tight bounds for Spearman’s footrule in the presence of missing data, with applications for independence testing.*
- **Zeng Y**, Adams NM, Bodenham DA. Scale two-sample testing with arbitrarily missing data. arXiv preprint arXiv:2509.20332. 2025 Sep 24.
 - *Perform two-sample scale testing in the presence of missing data without missing data assumption.*

OPEN SOURCE PACKAGES

- **Zeng Y**, Bodenham D, and Adams N. Package ‘wmwm’.
 - *R and Python packages for performing the Wilcoxon-Mann-Whitney two sample hypothesis testing in the presence of missing data.*
- **Zeng Y**. Package ‘bosfr’.
 - *R package for computing tight bounds of Spearman’s footrule in the presence of missing data.*
- **Zeng Y**. Package ‘abwm’.
 - *R package for performing the Ansari-Bradley two sample hypothesis testing in the presence of missing data.*

INVITED TALK

- StatML CDT White Board Talk Jan 2025
 - *Presentation title: Valid hypothesis testing with missing data*
- Imperial Internal Statistics Seminar June 2025
 - *Presentation title: Rank-based hypothesis testing in the presence of missing data*
- RSS Conference 2026 session on missing data (To be confirm)

SELECTED DATA SCIENCE PROJECTS

- [E-Commerce Order Cancellation Prediction](#)
 - *Predict order cancellation with highly imbalanced dataset for real-world impact.*
- [Deep Learning Hangman Game](#)
 - *Play word guessing games using deep learning with high success rate ($\sim 60\%$).*
- [Stock Clustering](#)
 - *Cluster stocks using daily return to produce balanced investment groups.*

SELECTED TEACHING EXPERIENCE

- Probability for Statistics (MSc) Summer term 2023
- Data Science (MSc) Spring term 2024
- Statistical Clinics (MSc) Summer term 2024
- Probability and Statistics (BSc) Spring term 2024

SKILLS

- Language: Chinese (Native), English (Proficient).
- Programming & Tools: Python, R, TensorFlow, pandas, NumPy, scikit-learn, MATLAB; Azure ML; HPC; Linux; Git.